

AI Resume Analyzer using NLP Techniques

Aman Chaudhary(Roll Number: 23035010651)

Trimester 7

B.Sc. (Honours) Data Science and Artificial Intelligence
Indian Institute of Technology Guwahati, India
aman.chaudhary@op.iitg.ac.in

Abstract

In this paper, we discuss the design and development of a resume analyzer that utilizes Artificial Intelligence algorithms with the help of Natural Language Processing techniques. The goal of this project is to use the resume analyzer algorithm to compare resumes against job postings and calculate their similarity scores.

To perform the task, the algorithm first extracts texts from PDF resumes, preprocesses the data using techniques such as tokenization, lower casing, and stopwords removal, and uses TF-IDF vectorization to convert texts into numerical vectors. Cosine Similarity technique is employed to calculate similarity scores between resumes and job descriptions.

Our resume analyzer outputs resume matching percentage scores, highlights missing keywords, and provides suggestions for

Keywords

Natural Language Processing, Machine Learning, TF-IDF, Cosine Similarity, Resume Analysis, Streamlit

1 Introduction

Recruitment is an essential procedure in contemporary organizations. For each job position, the organization receives hundreds of resumes, making resume screening difficult for recruiters to accomplish manually.

Manually assessing and screening resumes may lead to inconsistencies due to human errors and subjectivity of human judgments.

In order to mitigate these issues, companies resort to ATS systems to help screen resumes.

This paper presents a research aimed at implementing an AI-powered Resume Analyzer based on NLP algorithms.

2 Problem Statement and Objectives

2.1 Problem Statement

Classic resume selection techniques become ineffective as more resumes are submitted by applicants to an organization.

The problem that will be solved through this project is automation of resume evaluation and candidate-job matching through NLP technology.

2.2 Objectives

1. For resume screening using natural language processing techniques
2. For comparison between resumes and job descriptions
3. For computing similarity using TF-IDF and cosine similarity scores
4. For identifying gaps in keyword usage in the resume
5. For providing recommendations for making resumes ATS friendly

3 Methodology / Approach

3.1 Overall Workflow

The system follows a multi-stage NLP pipeline for resume analysis.

The workflow includes:

- Resume Upload
- Text Extraction
- Text Preprocessing
- TF-IDF Vectorization
- Cosine Similarity Calculation
- Match Score Generation

3.2 Technical Details

The following NLP techniques are used:

- Tokenization

- Stopword Removal
- Lowercasing
- TF-IDF Vectorization
- Cosine Similarity

TF-IDF is used to convert textual information into weighted numerical vectors.

$$TFIDF(t, d) = TF(t, d) \times IDF(t) \quad (1)$$

Cosine similarity is calculated using:

$$Similarity = \frac{A \cdot B}{||A|| ||B||} \quad (2)$$

4 Experiments and Results

4.1 Experimental Setup

Software Environment:

- Python 3
- Streamlit
- Scikit-learn
- PyPDF2
- NLTK

Hardware Environment:

- Apple MacBook Air
- macOS

4.2 Results

Similarity scores were correctly calculated for resumes and job descriptions.

Important findings are:

- Higher similarity scores were observed for greater keyword matching
- Skills missing from resumes were detected accurately
- Resume processing was very quick (a few seconds)

5 Discussion

This project showcases the use of Natural Language Processing for automating resume screening through automation.

TF-IDF is highly effective at recognizing significant terms, whereas cosine similarity offers valuable measures of similarity between resumes and job descriptions.

But the present algorithm relies primarily on keywords and lacks semantic understanding.

6 Conclusion and Future Work

This project has been successful in designing a Resume Analyzer AI system through NLP approaches.

The tool performs automatic resume analysis, evaluates similarity scores, recognizes keyword omissions, and provides recommendations.

Further development could involve:

- Implementation of transformers like BERT
- Semantic comprehension via Large Language Models
- Cloud hosting and database connectivity

7 Artifacts and Demonstrations

- **GitHub Repository:** <https://github.com/0123amanchaudhary-sketch/AI-Resume-Analyzer/blob/0123amanchaudhary-sketch-patch-1/README.md>
- **Video Demonstration:** https://youtu.be/NZ_mtDeRVFk
- **Technology Stack:** Python, Streamlit, NLP

References

- [1] Steven Bird, Ewan Klein, and Edward Loper, *Natural Language Processing with Python*, O'Reilly Media, 2009.
- [2] Scikit-learn Documentation, <https://scikit-learn.org>
- [3] Streamlit Documentation, <https://streamlit.io>